

7 The curve C has equation $y = \frac{x^2 - x - 3}{1 + x - x^2}$.

(a) Find the equations of the asymptotes of C . [2]

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(b) Find the coordinates of any stationary points on C . [3]

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- (c) Sketch C , stating the coordinates of the intersections with the axes.

[3]

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- (d) Sketch the curve with equation $y = \left| \frac{x^2 - x - 3}{1 + x - x^2} \right|$ and find in exact form the set of values of x for which $\left| \frac{x^2 - x - 3}{1 + x - x^2} \right| < 3$.

[6]